



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology

Academic Year 2024 - 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT

Course Code & Title: CS3591 & Computer Networks

Name of the Faculty member (s): Dr. K. Palraj, ASP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / Error Control
- **Course Outcome:** CO5
- **Topic Learning Outcome:** Pr09, 5b
- **Activity Chosen:** One Minute Paper

- **Justification:**

The **One-Minute Paper** is an innovative practice for error control that actively engages students in error detection, provides timely feedback, and fosters reflection and ownership over their learning. By utilizing this simple yet effective tool, educators can create a dynamic learning environment where errors are identified early and corrected promptly, ultimately leading to deeper understanding and improved academic performance.

- **Time Allotted for the Activity:** 10 Minutes

- **Details of the Implementation:**

At the end of the lecture, students were instructed to write about the error control concept that had been discussed. During the first 35 minutes, the instructor explained about the error control in the classroom. The students stated their understanding of the topic as well as the information for which they require additional explanation. Figure 1 and 2 shows error control in data link layer that students were wrote in the paper. This task assesses students' understanding of the idea of error control. This activity development the involvement of students in the class. This approach ensures that errors are not just identified but actively corrected, contributing to a more dynamic and effective learning experience.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO5	3	3	2	2	3	1	2	2	1	2	1	1	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9 & P10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	2	2	3	1	2	1	2	1	1	3
Justification for correlation	The students could be able to use the basic mathematical knowledge to build Data link layer	The students could be able to formulate and analyze physical layer for real-time applications	The students could be able to design the various data link layer functions for engineering problems and real time problems	The students could be able to interpret and analyze the real world application using transmission media	The students could be able to use various error control techniques	The students will complete the assessments in an ethical way	The students will work as a team and as an individual and demonstrate communication, problem-solving skills	The students will be able to identify the task required to build physical layer	The student will identify the technological advances made in the past and update the concepts in data link layer	The students could be able to apply computing skills to the data link layer solution to any real-world networking problems	The students could be able to provide the IT solution with the combination of data link and physical layer.	The students could be able to develop software project related to data link layer

- Images / Screenshot of the practice:

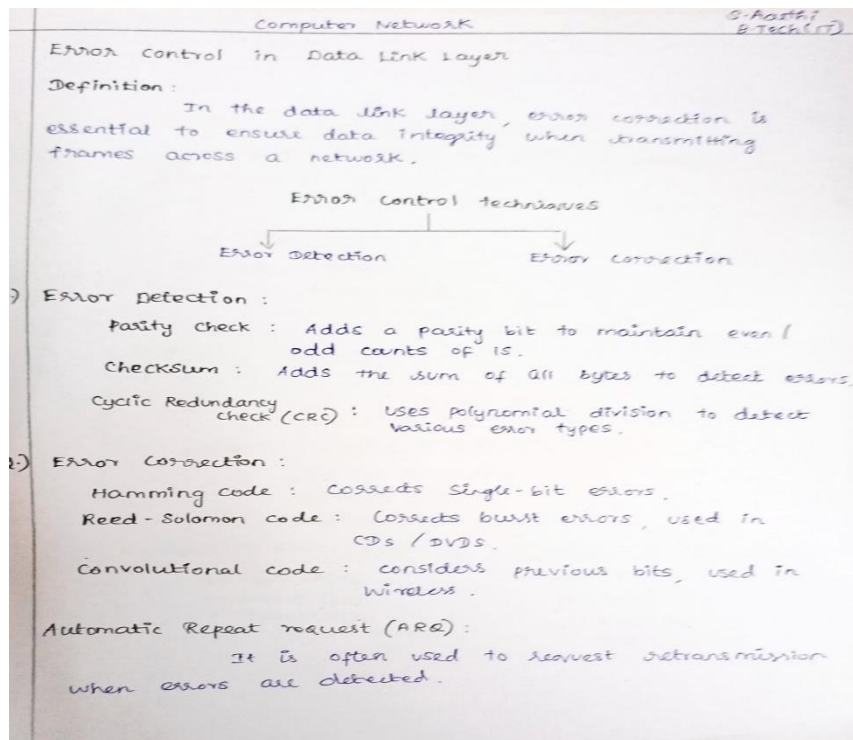


Figure 1. Sample One-Minute Paper of S. Aarthi

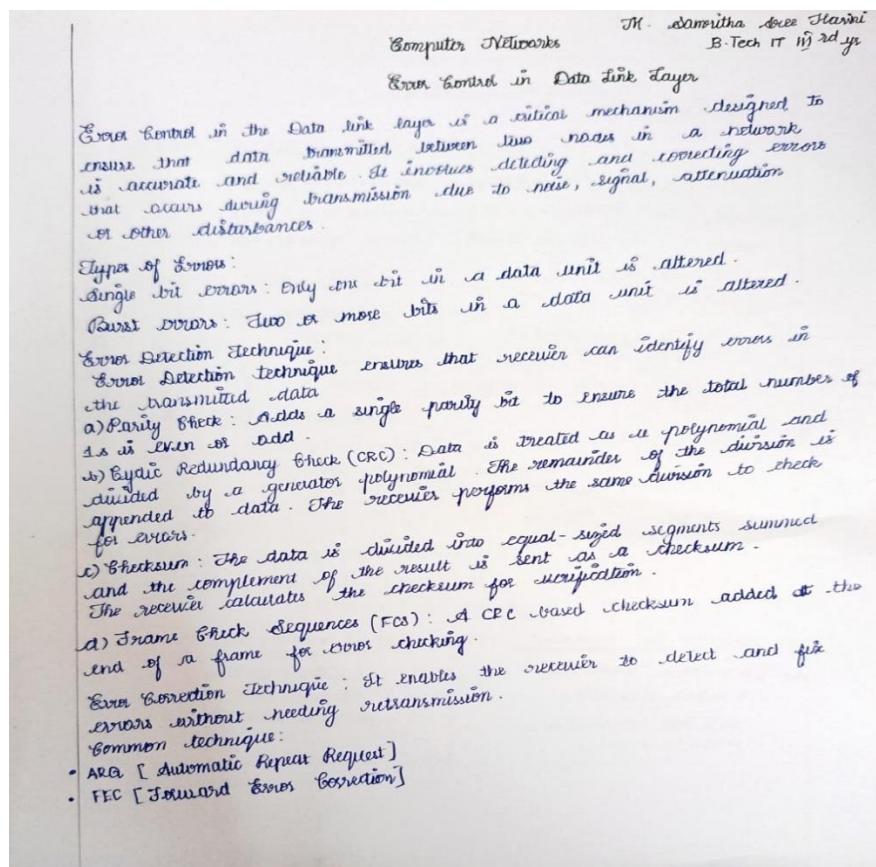


Figure 2 Sample One-Minute Paper of M. Samritha Sree Harini

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- The students felt that they can able to understand the events topic clearly after conducting this activity.
- They felt this activity makes them to concentrate and memorize the concept earlier.
- They are easily wrote about the topic for better understanding.

❖ *Benefit of the practice:*

- Students were actively participated in this activity.
- From this activity, the students can get more clarity in the error control topic.
- Students can able to write the topic without any confusion.
- Make the students to know the impact and importance of writing their views and understanding related to the topic and made them involve in the activity.

❖ *Challenges faced in implementation:*

- Some students are lack to write the concepts that they learnt in the class.
- Some students are unable to understanding the error control. After this activity that type of students , they got clear understanding.

References:

- ❖ <https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html>
- ❖ <https://www.unl.edu/gradstudies/current/teaching/minute>
- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/GM_CS3491_One_Minute_Paper.pdf
- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>

Signature of Faculty Member

HOD